



Artificial intelligence bias in medical system designs: a systematic review

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Abstract

Inherent bias in the artificial intelligence (AI)-model brings inaccuracies and variabilities during clinical deployment of the model. It is challenging to recognize the source of bias in AI-model due to variations in datasets and black box nature of system design. Additionally, there is no distinct process to identify the potential source of bias in the AI-model. To the best of our knowledge, this is the first review of its kind that addresses the bias in AI-model by categorizing 48 studies into three classes, namely, point-based, image-based, and hybrid-based AI-models. Selection strategy using PRISMA is adopted to select the 72 crucial AI studies for identifying bias in AI models. Using the three classes, bias is identified in these studies based on 44 critical AI attributes. Bias in the AI-models is computed by analytical, butterfly, and ranking-based bias models. These bias models were evaluated using two experts and compared using variability analysis. AI-studies that lacked sufficient AI-attributes are more prone to risk-of-bias (RoB) in all three classes. Studies with high RoB loses fins in the butterfly model. It has been analyzed that the majority of the studies in healthcare suffer from data bias and algorithmic bias due to incomplete specifications mentioned in the design protocol and weak AI design exploited for prediction.

Keywords Data bias · Algorithmic bias · Bias visualization · Bias accountability · Mitigating bias · Legal manifestations

SN Acronym/Symbol Description

- | | |
|----------|-----------------------------------|
| 1 ACC | Accuracy |
| 2 ANOVA | Analysis of variance |
| 3 ANCOVA | Analysis of co-variance |
| 4 AI | Artificial intelligence |
| 5 AUC | Area under the curve |
| 6 AUPRC | Area under precision-recall curve |

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7 BMI	Body mass index
8 BI	Brain imaging
9 CMR	Cardiac magnetic resonance
10 CVD	Cardiac vascular disease
11 CKD	Chronic kidney disease
12 CXR	Chest X-ray
13 CT	Computed tomography
14 CAD	Computer-aided diagnosis
15 <i>CI</i>	Confidence Index
16 COI	Coronary imaging
17 COVID-19	Coronavirus disease 2019
18 COMPAS	Correctional Offender Management Profiling for Alternative Sanction
19 CHARMS	Critical appraisal and data extraction for systematic reviews of prediction modelling studies
20 DL	Deep learning
21 EHR	Electronic health record
22 ECG	Electrocardiography
23 GDPR	General Data Protection Regulation
24 HDL	Hybrid Deep learning
25 HD-ai	Hybrid data-based AI-model
26 ID-ai	Image data-based AI-model
27 LVEF	Left ventricular eject fraction
28 ML	Machine learning
29 MRI	Magnetic resonance imaging
30 MCC	Matthew's correlation coefficient
31 MSE	Mean squared error
32 PD-ai	Point data-based AI-model
33 PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analysis
34 RT-PCR	Reverse transcription polymerase chain reaction
35 RoB	Risk of Bias
36 ROBINS-I	Risk of Bias In Non-randomized Studies - of Interventions
37 SEN	Sensitivity
38 SI	Stroke imaging
39 SPE	Specificity
40 USS	Ultrasound
41 WHO	World health organization

1 Introduction

Artificial Intelligence (AI) has gained momentum in the (a) field of healthcare, which includes several fields of view such as cardiology, urology, angiography, endocrinology, neurology [83, 120], (b) face recognition [149], (c) police interrogations [95], (d) text classification [134], (e) smart city development [42] and many more [68, 113, 128]. Over the years, AI-based predictive models and decision-making systems have raised concern about the negative amplification in the outcome due to AI-bias. Generally, bias can be defined in the outcome of an AI-based predictive model where the results are identified to be inclined towards a particular group of marginalized people based on their race,

gender, ethnicity, society, and kinship [29, 75, 102]. In the healthcare industry, bias in AI-based prediction models can be introduced due to incorrect data assessment, model limitations, small training size, less community participation, and other unseen factors [39]. This restricts the clinical deployment of AI-based prediction models and emphasizes the need to analyze the importance of outcomes from paper to practice [99]. Hence, an analysis of risk-of-bias (RoB) in the healthcare industry is imperative to provide unbiased, accurate, and efficient diagnosis and prognosis to the population. In addition, predictive results should be analyzed for bias accountability at an early stage to save implementation and financial efforts. Recently, several works have been proposed to analyze and reduce the RoB in AI in healthcare [33, 138, 142]. These works analyzed bias theoretically, measured it statistically and assessed it through quality tools [22, 27, 44, 116, 132]. To the best of our knowledge, no work has qualitatively and quantitatively analyzed the bias in AI-based prediction models in the healthcare domain.

Impact of bias can be qualitatively computed by non-randomized AI strategy tools such as PROBAST [157] and ROBINS-I [136]. Quantitative assessment of bias includes the computation of statistical measures namely, accuracy (ACC), sensitivity (SEN), specificity (SPE), p-value, F1-score, precision, recall, and many more. In addition, RoB can also be quantified with the help of quality and fairness assessment toolkits such as aequitas [121], BIAS [82], critical questions tool [152], Cochrane tool [84] and many more [71]. These methods study the various bias attributes such as explainability, interpretability, robustness, reliability, generalizability, interoperability, and accountability [15, 98]. These attributes are helpful in visualizing the bias at various stages of the AI-model. Historical biases, societal bias, sampling bias, representation bias, human bias, training bias, and measurement bias are essential to visualize the unbiased and accurate outcome of the predictive model. Also, disparity in the dataset collection, measurement of a particular feature, sample size of the dataset, and the biased evaluation metric need to be visualized for the bias accountability in the model. In general, bias flow has broadly five components which include the potential sources of bias such as (i) data bias, (ii) algorithmic bias, (iii) bias evaluation, (iv) bias mitigation, and (v) bias legal manifestations. Each component has three categories to represent and elaborate bias, the related details are illustrated in Figs. 1 and 2. These figures demonstrate the significant levels of bias study in AI-model exploited in the medical field at the microscopic level along with bias evaluation and bias mitigation strategies with legal considerations.

The key innovations in this study are to analyze bias in AI models by classifying them into three classes based on the exploited input data. We hypothesized that AI studies in each class with weak design system are more biased than those which have strong design architecture. In addition, the unbalanced datasets and quality of datasets utilized for training the model impacts the generalizability and interoperability of the model. It is due to this reason that AI-studies in healthcare are not covering all the facets of AI design architecture crucial for the unbiased outcomes as well as clinical deployment. The main contributions of this study are as follows:

1. *Classification of the data sets*: Three classes of datasets, namely Point, Image, and Hybrid data sets for studying the bias characteristics and four types of attribute clusters namely, demographics, architecture, data, and clinical evaluation comprising of 44 crucial AI attributes designed for bias analysis.
2. *Types of the bias models for analyzing the AI studies*: Butterfly Model and Analytical Model for comparison of bias between different AI-studies. Each study is evaluated for

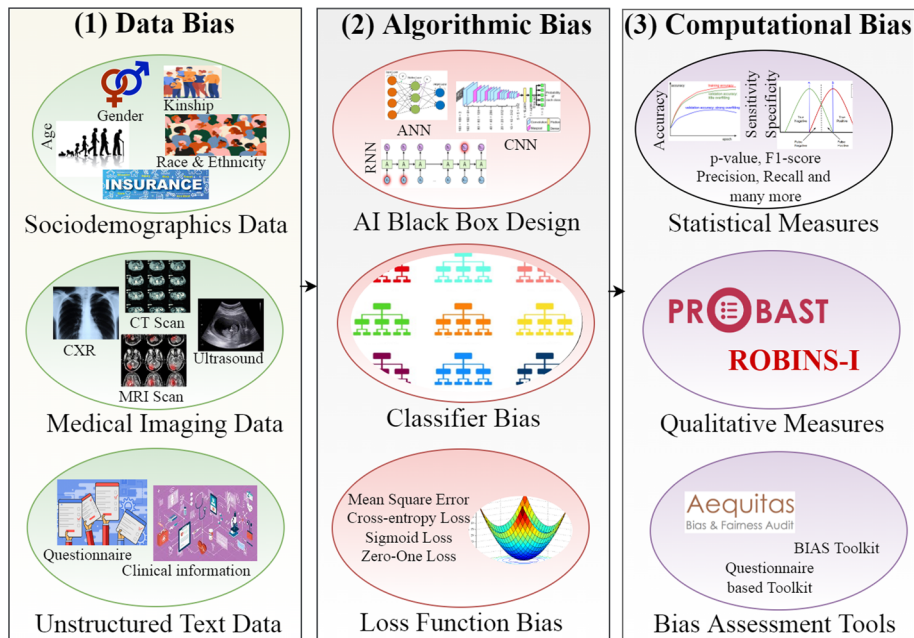


Fig. 1 Systematic representation of various components of bias-flow in AI-based prediction models. (Contd...)

- radial-bias map, regional-bias area, ranking-bias score, and analytical method to categorize into three RoB zones namely, *high-risk*, *moderate-risk*, and *low-risk*. In addition, variability study is also performed to check the agreement between the two AI experts.
3. *Correlation between the four bias estimation methods*: Correlation between four bias estimation techniques namely, Radial, Areal, Ranking, and Analytical Method is computed using Venn diagram. Studies that fall under the overlap region of the four methods inferred the strong relationship between the bias estimation methods.
 4. Bias mitigation strategies will be assessed to reduce the impact of bias and legal regulations will be reviewed for effective implementation and to enforce transparency in AI-model.

The remainder of the paper is organized as follows. In Section 2, search strategy using PRISMA model is presented for selecting the research studies for this work. Section 3 discusses the bias life cycle and various bias classes based on the input data classification along with their salient features, similarities, and differences. The innovative bias computation in each study under each class using butterfly model, ranking bias score and analytical model is depicted in Section 4. In addition, variability study is elaborated to compare the results from experts. Section 5 shows the bias mitigation approaches along with the legal implications due to bias. Critical discussion and the recommendations/challenges for bias in AI-model are discussed in Section 6. Finally, the concluding remarks are sketched in Section 7.

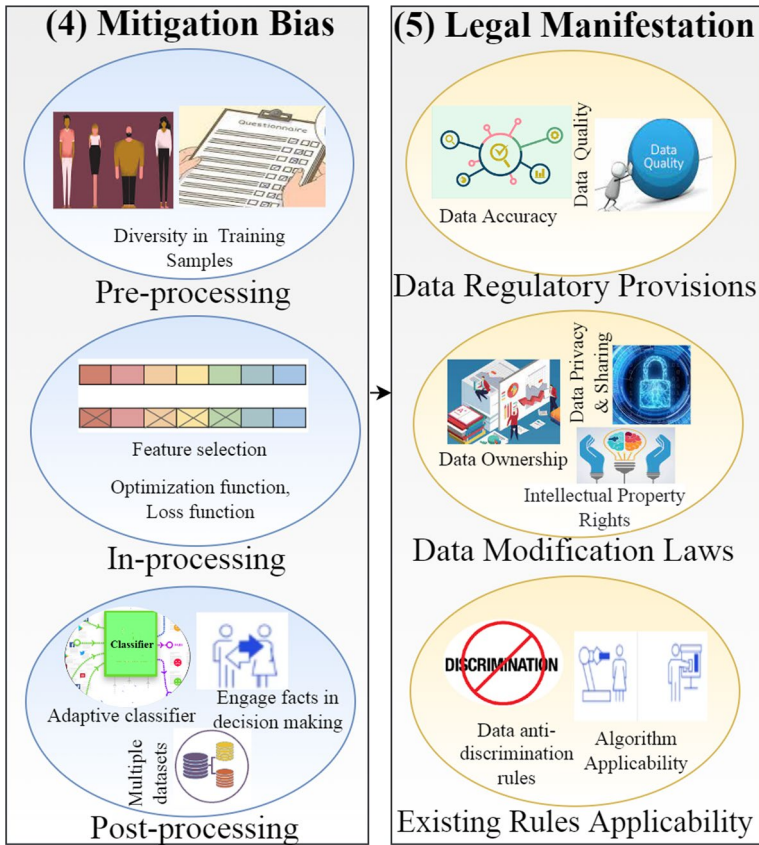


Fig. 2 Systematic representation of various components of bias-flow in AI-based prediction models

2 Search strategy using PRISMA model

A thorough search was conducted for searching the most relevant articles using IEEE, PubMed, Google Scholar and other sources (ArXiv, ResearchGate etc.). We did not explore the Cochran database since it was engineering oriented. The keywords used for searching the articles were “Bias in AI”, “Managing Bias in AI”, “Identifying Bias in AI”, “Bias in AI in medical Imaging”, and “Addressing Bias in AI in Medical Imaging, data bias, demographics bias, algorithmic bias, bias mitigation, bias distribution. in software engineering, bias liabilities”. Figure 3 illustrates the PRISMA model consisting of the selection criteria for the related studies discussed in this work. Based on the above-mentioned search methodology, a total of 4,604 studies were identified. A total of 2,214 duplicate articles were located using the “Find Duplicate” function of EndNote software of Clarivate Analytics and subsequently filtered out from the analysis. Further, three exclusion criteria namely, E1, E2, and E3 excluded the number of articles 1460, 556, and 126, respectively from the analysis. E1 criteria excluded the records that did not utilize the AI techniques such as hybrid deep learning, deep learning or machine learning in their prediction architecture. E2 excluded the non-relevant articles which were not related to the domain. However, the articles which have insufficient details in the abstract, datasets and discussion were excluded using criteria E3. Finally, a total of 72 most relevant articles were selected for the study.

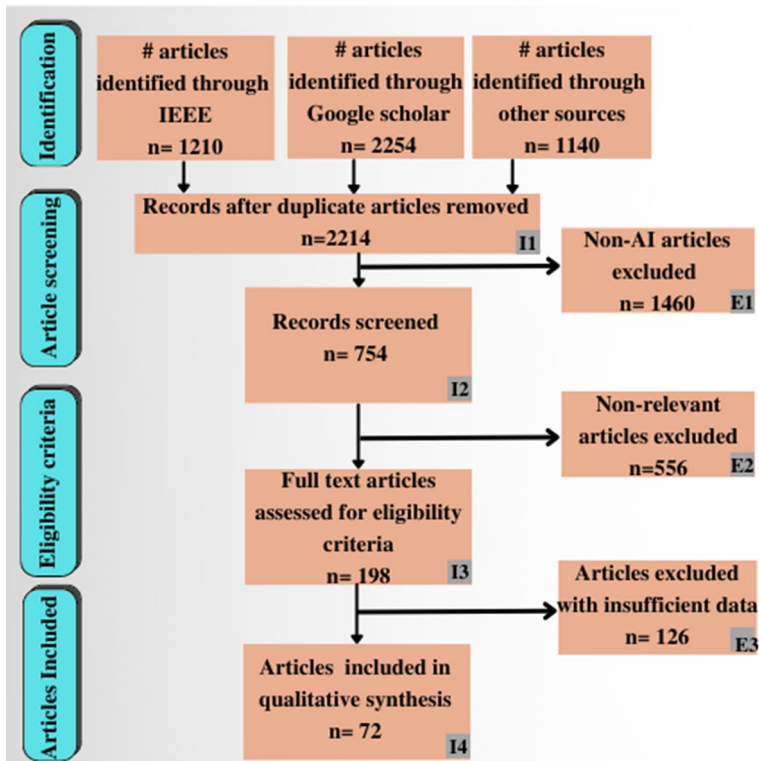


Fig. 3 Study selection using the PRISMA model

2.1 Hypothesis and risk-of-bias acceptability criteria

We hypothesize that the AI-study in healthcare is at high RoB due to missing patient's sociodemographics, weak algorithmic design, incomplete datasets, and insufficient evaluation metrics. The four acceptability criteria namely, Radial Bias Mean (RBM), Regional Bias Area (RBA), Ranking Bias Score (RBS) and Analytical Evaluation (ANAL) are considered to categorize in AI-studies based on their strength in three zones (low, moderate and high). For all the AI-studies, mean scores of all the considered crucial AI attributes should be $\geq 80\%$ to be *low RoB*. The study in the *high-risk* of bias will lose fins in the butterfly model and is not suitable for clinical practices [142]. The cumulative scores are computed for each AI-study and two cut-offs namely, *low-moderate* (LM) and *moderate-high* (MH) are computed to identify the RoB. Studies with LM cutoff (< 2.5) are under *low-bias* zone.

3 Bias life cycle

Bias in AI-based clinical decision-making systems can lead to unseen classifications and results. This emphasizes the need to understand the impact of bias at various stages of AI-model [156]. The AI model should be analyzed for transparency, opaqueness, and fairness before the deployment and/or product delivery. Figure 4 illustrates various stages of the bias life cycle and its visualization in AI-based predictive model.

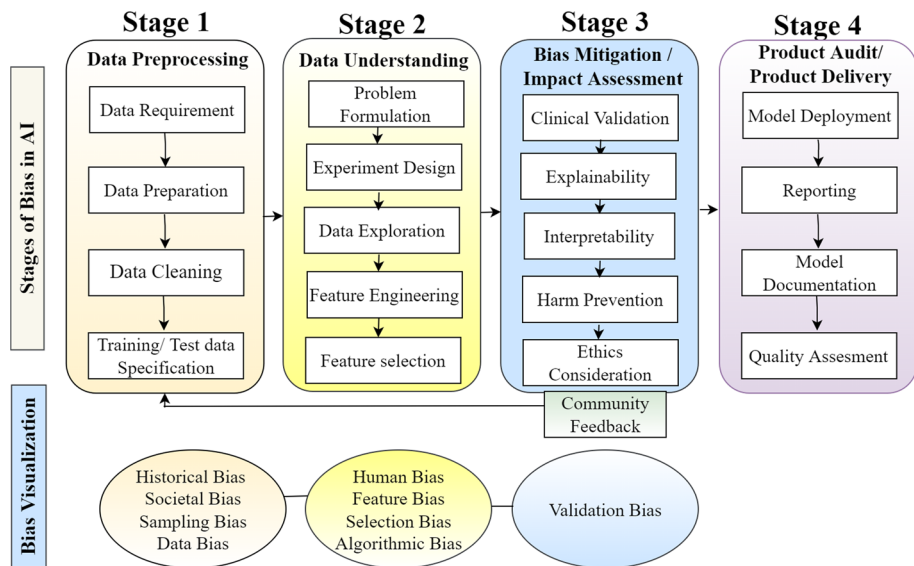


Fig. 4 Various stages of *bias life cycle* and its visualization

Stage I of the data pre-processing step (vertically down) includes data collection, data preparation, data cleaning and training data specification. Limitations and negligence during data collection and gathering are responsible for data bias in the model [47]. Historical bias is inductive in data due to skewedness towards a marginalized group. Societal bias is embedded in the data due to barriers that pre-exist in society. This will lead to imbalance and discrimination in the datasets [44]. In addition, data collection from multiple alternate sources is also responsible for inducing bias in the training data [18]. Models trained on different public datasets generate biased outcomes as the process of data collection and processing is hidden [129]. Unawareness and ignorance of sociodemographic information namely, gender, age, ethnicity, race, socioeconomic information and clinical information such as medical history, during training and test data specification are accountable for misclassification and biased outcome. These biases exacerbate during AI model implementation as these models are highly data-dependent. The stage II of data understanding deals with bias introduced during problem formulation, algorithmic design, data exploration and feature selection. The bias in problem formulation for the predictive model is introduced due to the limited domain knowledge of the developer. Automated and black box design of the AI-based predictive model is responsible for algorithmic bias. Data exploration that includes the generation of synthetic data, surrogate data and data manipulation for model training is responsible for selection bias. In addition, embedded bias in selected features for model training is liable for feature bias [35]. Sometimes irrelevant correlations between the selected and sensitive non-selected features lead to incorrect predictions due to bias which is hard to quantify [117].

Next, stage III of the bias life cycle deals with bias mitigation strategies mandatory to reduce its impact before the final product delivery. The model diagnostics namely, explainability, interpretability, transparency, generalizability and interoperability are substantial for deployment of AI-model. Bias accountability can be analyzed for explainability and interpretability to explain the black box design of the AI-model global logic [51]. Data

transparency is necessary to consider the data-related ethics namely, accuracy, quality, sharing, and privacy [72]. Generalizability and interoperability of AI-model are imperative for global acceptance of the predictive model. Model evaluation for validation bias is required for harm prevention and to mitigate its adverse impact on human beings after clinical deployment [76].

An important step to mitigate the impact of bias in the outcome of AI-model is the community feedback loop. This step emphasizes the need for bias awareness data collection to improve the performance of predictive model. Critical aspects in the datasets such as completeness, diversity, discrimination, and transparency can be evaluated for unbiased algorithmic design. Input-sensitive features shall be identified and evaluated to mitigate the impact of individual samples on model prediction. Hence, feedback loops ensure that the negative influence of the bias can be minimized or eliminated for improvement in the model outcome. Finally, model documentation and reporting are done to facilitate the prediction model's technical applicability, and clinical deployment from paper to practice [99]. The audit process can be documented to prepare reporting template for model quality assessment for deployment into clinical practices [100]. The next section will discuss the input data-based classification of AI bias into three classes.

3.1 Three types of bias

AI-based prediction models are increasingly progressive in the healthcare industry. Automated AI-based decision-making systems are in high demand due to their outcome for early detection of chronic diseases. Recently, it has been evident that the AI-based healthcare the prediction model is biased in terms of prediction and raising concern for their clinical deployment. Primarily, bias in the AI-model can be observed through the type of input data exploited and the data gaps during data collection [41]. AI-based algorithms learn and train from historical data that encodes the bias in model. This may be either due to lack of diversity in data collection methods or limited knowledge of the AI-model developer [35]. Hence, it is crucial to categorize the work based on the input data to better understand and analyze bias.

In order to analyze the biases in AI models, we have categorized the work in three classes namely, point data-based AI model (PD-ai), image data-based AI model (ID-ai) and hybrid data-based AI model (HD-ai). The fundamental difference between these three clusters is in the input data type utilized for analysis. Unstructured information such as gender, sex, ethnicity, body mass index (BMI), medical history, and medical prescription is utilized as the input for outcome prediction in PD-ai models [92, 111]. Medical imaging such as chest X-ray (CXR), magnetic resonance imaging (MRI), ultrasound scan (USS), and computed tomography (CT) are exploited as input in ID-ai models [81, 116, 148]. HD-ai models prefer to utilize both unstructured text information and medical imaging data for accurate and unbiased analysis [115, 154]. In addition, HD-ai model uses imaging data labeled for multiple diseases to analyze for algorithmic bias. The salient features of the three AI-bias classes are tabulated in Table 1. The details related to data analysis, feature extraction, performance, computational, and many more for AI-bias classes are compared in this table. In order to differentiate between the three classes based on data standards, model development and statistical measures, the details are tabulated in Table 2. Table 3 illustrates the differences between the three classes of bias in detail based on the substantial parameters namely, model representation, input data type, classification, segmentation [37], and privacy. In addition, it is substantial to study the characteristics of bias and compute its impact in these

Table 1 Salient features of three classes of AI-bias

SN	Attributes	PD-ai	ID-ai	HD-ai
1	Data Requirement	Low	Moderate	High
2	Data Availability	Low	Moderate	High
3	Data Handling	Structured or unstructured patient data	Patient's medical imaging data	Structured or unstructured data with patient's imaging data
4	Feature extraction	Limited	Manually or automated	Manually and automated
5	Advantages	Can identify individual sample limitations	Early diagnosis of critical diseases	Relatively Better outcome in terms of early prediction of disease
6	Performance	Low	Moderate to high	Relatively high accuracy
7	Limitations	Less accurate	Data quality and sample size	Availability of both input data
8	Computational Complexity	Less	Moderate	High
9	Hardware bias	NA	Different devices for generating medical imaging data	Different devices for generating imaging data

Table 2 Similarities between the three classes of AI-bias

SN	Attributes	PD-ai	ID-ai	HD-ai
1	Explainable AI models	✓	✓	✓
2	Discrimination-aware AI models	✓	✓	✓
3	Data standards and security	✓	✓	✓
4	Multiple input datasets	✓	✓	✓
5	Model generalizability	Limited	Moderate	Relatively high
6	Statistical measures	Limited	✓	✓
7	Similarity measures	×	Analysis of outcome	Analysis of outcome
8	Types of attributes	Protected, unprotected and exploratory attributes	Protected, unprotected and exploratory attributes	Protected, unprotected and exploratory attributes

three categories to mitigate the effect in medical diagnosis, facial recognition, textual classification, and psychiatric analysis. Hence, the applications of three bias classes in various domains are tabulated in Table 4.

3.1.1 Bias attributes details under PD-ai

Mostly studies under PD-ai are survey-scenario, questionnaire, and diagnosis reports based retrospectively or prospectively defining the reason of bias in the outcome [40, 108, 118]. These models primarily focus on identifying the potential biases due to data collection gaps in electronic health records (EHR) that include admission forms, test reports, procedures, and medication details. The information gap related to patient's biomarkers such as voice, physical indicators, lifestyle, and climatic conditions should be incorporated in the AI-model for accurate outcomes [109]. PD-ai model analyzes the bias in clinical studies due to missing data, underestimation, underdiagnosis, and individual sample problem that may arise due to neglecting substantial parameters during questioning and health data records collection [49]. The questionnaire-based healthcare app suffered from racial inequalities as it suggested different diagnoses for different race of people keeping all other parameters identical [101, 108]. Bias exists in PD-ai prediction model due to societal and representation barriers that arising from geographical diversity in data collection [15]. In addition, patient characteristics, structured clinical prescription, and diagnosis might be ignored while estimating the prediction models [131]. Hence, these models predict the false positive outcomes on specific datasets and the implication to clinical models is not fundamentally possible. The bias attributes details of the representative work under PD-ai are tabulated in Appendix (Table 6). The significant parameters for bias visualization, accountability, and computation in work are extracted to identify the potential source of bias in the outcome.

3.1.2 Bias attributes details under ID-ai

Artificial Intelligence has started to penetrate the field of medical imaging at a very warp speed in different medical domains such as brain imaging (BI) [147], skin cancer [130], breast cancer imaging [146], liver cancer imaging [69], atherosclerosis imaging and plaque

Table 3 Differences between the three classes of AI-bias

SN	Attributes	PD-ai	ID-ai	HD-ai
1	AI Model	Mostly ML-based model	ML-based or DL-based model	ML, DL, or HDL based model
2	Algorithm types	Survey-based scenario algorithm	Feature extraction and sample selection-based algorithms	Features selection and extraction from imaging with demographic and other clinical details
3	Model representation	Structured or unstructured text data	Image data	Structured or unstructured text with imaging data
4	Data segmentation	×	Segmentation of image based on clinical diagnosis	Segmentation of image with augmentation of additional information
5	Data Classification	Classification based on the extracted information	Classification based on clinical and non-clinical outcomes	Classification of outcome based on labeled clinical and non-clinical data
6	Data Privacy	Patient’s consent for data usage	Radiology Image usage consent from the organization	Both patient and radiologist consent for using data

Table 4 Application areas for three types of AI-Bias

SN	Attributes	PD-ai	ID-ai	HD-ai
1	Medical domain	RT-PCR report for covid-19 severity detection [40]	CXR, CT, and MRI for covid-19 detection, pneumonia prediction, and Alzheimer [141, 143]	Additional information with imaging for personalized treatment for Alzheimer, leukemia etc. [99]
2	Societal domain	Criminal interrogation [95] and recruitment AI models [5, 70]	Identify bias in the prediction of covid-19 in data from different sources [23]	Facial detection using demographic information with kinship data [48]
3	Textual classification	Bias in online language translators [133]	Not independently exists	Not independently exists
4	Face analysis	×	Face recognition [85]	Face detection and recognition with ethnicity and race information [48]
5	Psychological analysis	Psychological issues with low socioeconomic groups [49]	Brain MRI imaging segmentation for impacted areas [112]	Combining both informations for more accurate analysis [34]
6	Legal issues	×	Addressed in terms of data security, IPR, and privacy [43, 98]	Addressed in terms of data quality and data discrimination

characterization [2], coronary imaging (CI) [9], COVID-19 imaging and lesions classification [145], stroke imaging (SI) [30] and many more [65–67]. These techniques have evolved from conventional to non AI-based modeling paradigms [36].

Medical imaging modality such as CXR, cardiac magnetic resonance (CMR), CT, USS [122], and many more are widely used for detecting and tracking severe diseases [23–25, 78, 112]. ID-ai models utilize the publicly available image datasets of these modalities to predict severe diseases [56]. Bias in clinical AI-model can be introduced either by the biased features that specify the disease or the hardware module used for image data collection [43, 150]. The automatic feature extraction module cannot be generic in application to the societal imbalance subpopulation. In addition, the usage of specific devices to capture imaging details of a critical patient that a suspicious one can be reason for bias in the model [135]. In-built bias in clinical imaging collection can be introduced due to calibration differences in the hardware modules [106]. Bias in imaging data collection can be noticed by the difference in the outcomes of the prediction model while dealing with different subpopulation characteristics. The quality and quantity of data are paramount for AI-based clinical models. The data accuracy, security, privacy, and ownership should be designed to address the lack in the regulation of the model design. Automatic feature extraction process and data quality should be continuously monitored to restrict the negative impact of bias on the AI clinical models. The data extraction and storage policies should be standardized to hinder propagation bias in the AI-models [90]. Details of bias attributes under ID-ai class are given in Appendix (Table 7). Significant parameters for bias computation, visualization, accountability, and data source are tabulated to investigate the possible reason for bias in ID-ai class of AI-model.

3.1.3 Bias attributes details under HD-ai

The healthcare industries are adapting AI-based prediction model for diagnosis, prognosis [57], and treatment [124] of many critical diseases. However, the accuracy of these models is the foremost concern in providing the unbiased and fair decision system for healthcare. For this, hybrid-data based AI model has gained momentum in the healthcare industry. To provide effective and accurate medical outcomes, medical imaging modalities are integrated with patient's structured information while training the AI models [73, 100, 127]. The training of prediction model diagnosis comprising datasets of medical imaging such as CT scan, ultrasound scan, CXR, and MRI for diagnosis of critical diseases such as cardiovascular diseases (CVD), chronic kidney disease (CKD), Alzheimer, and Leukemia is considered insufficient. To improve the performance of these models, the patient's demographic information such as gender, age, race and ethnicity should be incorporated into the training dataset. Unstructured set of questionnaires and the patient's medical history should also be analyzed before concluding the final outcome. HD-ai models aim to identify the sources of bias that restricts the generalized implementation of these models. These models try to categorize the under-diagnosis of a patient due to imbalanced datasets [73]. However, the performance of these models is limited by the less availability of sociodemographic, socioeconomic, physical characteristics, and psychological symptoms information with the medical imaging data. In addition, the data quality and security issues such as patient consent, and organization consent must be considered before the implication of these models into clinical practices. Appendix (Table 8) tabulates the bias attributes details under HD-ai class. The details are extracted to determine the source of bias in HD-ai work. The selected work under HD-ai class identifies the paramount attributes to conclude certain steps to mitigate bias in medical outcomes.

4 Bias computation models

In healthcare imaging, potential bias can be introduced in AI-based prediction models due to either data or algorithmic design [87, 105]. Data bias is associated with input dataset issues during production, mapping, and training. Models can include data bias due to insufficient training data, missing data, misclassification, synthetic, and surrogate data [118]. Data bias can also be embedded in the system due to data collection and generation methods i.e., difference in medical imaging quality due to different hardware devices [150]. However, algorithmic biases occur due to inefficient and weak designing of AI-based predictive models. If bias is introduced in the outcome due to the complex, adaptive and automated nature of system design even then also unbiased input data is considered to be algorithmic bias [13, 77]. Bias can also be introduced at various stages of AI-model i.e., pre-processing, in-processing, and post-processing [98]. Unbalanced datasets, data disparity, dataset incompleteness and dataset split into training, test, and validation lead to bias during pre-processing stage. Bias embedded in selected features, optimization functions, loss functions, and black box model designs are responsible for in-processing bias [131]. At the post-processing stage, bias can be introduced due to classifier design, decision boundaries, and calibration [98]. In AI-based medical systems, these biases lead to either incorrect prediction or underdiagnosis of medical outcomes [127]. In addition, the generalizability and interoperability of these models is substantial to evaluate with change in sociodemographic information, lifestyle and climatic environment. The next section will detail the data bias in AI-model.

4.1 Data bias

Data bias in AI-model can be deliberated either due to bias in demographics data or bias in data selection. Primarily, data bias can be introduced in the dataset during any type of data collection [47]. During data collection, little attention is given to include person's demographics details namely, gender, race, age, ethnicity, socioeconomic status, insurance, and kinship in the dataset. Sociodemographics inequalities in the dataset will lead to inconsistent outcomes for marginalized group [93]. The imbalanced data will result in specialized privileges to a specific group in the medical diagnosis [127]. Limited data sample size causes biased prediction due to impartiality and incompleteness in the dataset [64]. In addition, data bias can be embedded into the datasets due to different data collection methods, and devices. Difference in hardware configuration and quality of the devices used to capture datasets instigates variability in the collected samples, leading to negative bias [106].

Data selection also plays a vital role in exacerbating data bias in AI-model. Potential data cleaning steps such as data manipulation, synthetic data generation, surrogate data incorporation, corrupted and stale data analysis should be examined for biased outcomes in AI-based model. Data manipulation from alternate sources leads to inconsistency in the datasets [26]. The generation of synthetic data by scaling, rotating, and orientation to meet the requirement of the training sample data leads to biased results due to less diversified datasets [48]. Surrogate data such as the zip code used to determine the race of the patient should be adequately analyzed for its appropriate usage in the model [15]. The limitations of surrogate data should be reviewed for its participation in the input datasets. Corrupted and stale data from unreliable sources introduce bias in the outcome as the model learns

unfairly. The corrupted data may be incomplete and inaccurate that will lead to hard to quantify biased output.

4.2 Algorithmic bias

Bias in the outcome of AI-based predictive model is observed even if the input data is unbiased. System design and architecture are responsible for algorithmic bias in the model [7]. Reasons for algorithmic bias in the system are the black box nature of the AI design, choice of sensitive input features, bias due to classifier, loss function and optimizer, lack of context, and concept specifications in the development of the algorithmic design [105]. In addition, oversampling of training data, underrepresentation of population, and testing of a model on a non-specific data will automate bias in the outcome of predictive models [107].

Algorithmic bias in AI-model can be introduced in the outcome due to the missing training, validation and test specifications [32]. Generally, developers do not publicize the exact training, validation, and testing data split information. Model trained on less and/or missing data leads to underestimation and underdiagnosis [49, 127]. However, excessive trained model causes overfitting of the algorithm and prediction errors. Hence, the intended purpose, generalizability, and interoperability of the algorithms cannot be evaluated and generated biased outcome. Algorithmic bias can also be triggered through the inbuilt bias in the feature extraction and selection process. Feature extraction methods should be efficient enough to extract generic and discriminative features from the image. Deep learning methods exploited to extract features from medical images impacted the image quality and so the feature importance [79]. Sometimes feature distribution in medical imaging is non-uniform i.e., it is more informative for one gender in comparison to others. While feature selection, sensitive input features must be identified as their ignorance in the model design may impact the outcome. Also, non-availability of the sociodemographic information in the algorithmic design will generate biased predictions [101]. Another source of algorithmic bias is the black box nature of the AI-model design. The algorithmic processing design that includes the number of hidden layers, training weights, loss function, and classifier design exhibits significant bias in the prediction. Influence of the fundamental aspects of each element of AI-model is yet to be discovered by the developer and the designer. Interpreting certain predictions is impossible due to the AI-model's hidden design architecture. In addition, loss function such as mean square error introduces bias in the training weights impacting the image feature quality and affecting the prediction [94]. Also, classifiers could generate biased and unfair outcomes based on the input data. The classifier may lead to disparity in the results by classifying the same information into different subgroups [125].

In another line of work, lack of contextual standards and causal reasoning for the AI-model could lead to algorithmic bias. Usage of the algorithm for a non-specific and inappropriate context predicted unwanted results. In addition, causal assumptions are substantial for the understanding and fairness of the algorithm [80]. Casual inferences are counterfactual that ensure that the diagnosis decision should not be dependent on the specific gender. However, the implicit biases in the systemic standard of the algorithm lead to differential diagnosis to the patients having similar medical factors based on their sociodemographics [137].

4.3 Bias computational models

The computation of RoB in AI-model is essential before the transition of the system from paper to practice. AI-models need to be evaluated on critical AI attributes to ensure the clinical validation of the proposed work. In this work, we have proposed to evaluate RoB in a study using RBM, RBA, RBS, and ANAL. A study's strong and weak AI attributes are distributed in a radial span of 360 degrees to generate RBM. RBA is an indirect measure of RoB in a study generated by analyzing the area strength between the strong and weak AI-attributes. The crucial 44 AI-attributes for bias study in work are categorized into four clusters namely, sociodemographic cluster (C1), bias accountability cluster (C2), data bias cluster (C3), and bias evaluation cluster (C4). C1 cluster contains the societal information about the patient's (A1-A7) namely, (i) gender, (ii) race, (iii) age, (iv) ethnicity, (v) insurance, (vi) kinship, and (vii) clinical history. The second cluster (C2) consists of details about bias accountability related to AI algorithms (A8-A20) such as (i) type of AI architecture, (ii) feature extraction methodology, (iii) feature selection process, (iv) data augmentation, (v) robustness, (vi) reliability, (vii) explainability, (viii) interpretability, (ix) transparency, (x) generalizability (xi) interoperability (xii) discriminability, and (xiii) reporting. Third cluster (C3) includes (A21-A31) as (i) number of datasets in the study, (ii) size of the datasets, (iii) type of input data, (iv) historical information in the dataset, (v) synthetic data generation, (vi) data manipulation (vii) testing on the unseen dataset, (viii) data quality, (ix) data accuracy, (x) data privacy, and (xi) data security. The fourth and final cluster (C4) consists of evaluation parameters of bias in the model. This cluster has thirteen AI attributes (A32- A44) namely, (i) number of evaluation parameters used (ii) ACC, (iii) SEN, (vi) SPE, (v) F1 score, (vi) precision, (vii) ANOVA (viii) p-value, (ix) AUC)/AUCROC, (x) similarity score, (xi) recall, (xii) bias qualitative test for evaluation, and (xiii) risk analysis. The details of the grading scheme of these attributes are tabulated in Appendix (Table 9).

4.3.1 Radial bias map

To analyze the risk of bias in a study, the strength of 44 AI attributes is evaluated under four clusters. The crucial AI attributes are distributed as 7, 13, 11, 13 respectively under each cluster. The strength of AI attributes is evaluated and represented in 360-degree directions using the spokes and wheel model. Each spoke is estimated by multiplying attribute weight with the radius of the spoke. Attribute weights are determined by experienced AI professionals capable of identifying the contribution of each attribute toward bias. Figure 5a & b illustrates RBM for 48 studies (8×6 grids) in each bias class radius sorted in butterfly spline fitted curve for Expert 1 and Expert 2, respectively. The curve consists of four sectors and each sector corresponds to each cluster of AI attributes representing each fin of the butterfly model.

Then, the spoke length of each AI attribute is computed by multiplying the weight of AI attribute with 80% of image size (512). The sum of spoke length (τ_{C1} , τ_{C2} , τ_{C3} and τ_{C4}) is calculated for each cluster. Then, the sum of two top and two bottom clusters represented as τ_t and τ_b , respectively. Absolute error (δ) between (τ_t) and (τ_b) is then calculated. Finally, the normalized biased value for each AI-cluster is determined by dividing the absolute error (δ) with the total number of AI-attributes. The studies are arranged in the decreasing order of risk of bias. Studies at high RoB will lose fins and lie in the region

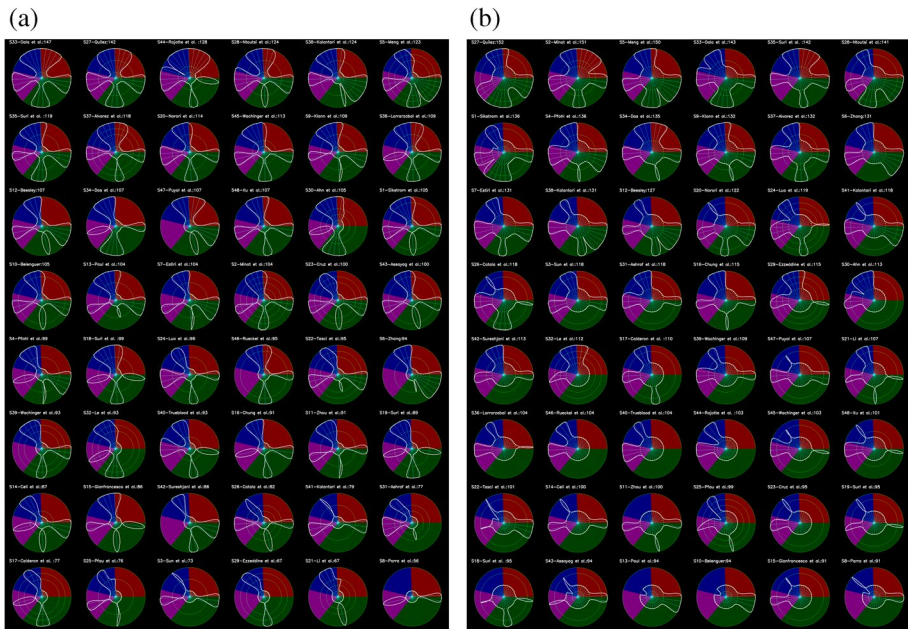


Fig. 5 Illustration of radial-bias Maps for 48 studies selected under each class for (a) Expert 1 (b) Expert 2

of high RoB. RBM of each study is named as: Sxx-Name:value where Sxx is the study number, name is author's last name in the study and value is the estimated normalized bias value for the study.

4.3.2 Regional bias area

We have utilized Regional Bias Area (RBA), an indirect method to compute bias in a study. Difference between the area of best and worst performing AI attributes determines it. The configurationally RBA map for the 48 selected studies in 8×6 grid for experts namely, Expert 1 and Expert 2 are presented in Fig. 6a & b, respectively. The studies are arranged in the order of increasing bias area. The white region depicts bias area in the figures. Studies having low RoB have less white region in comparison to studies with high RoB. Each study is named as: Sxx Name: value. Sxx represents the study number, Name is the last name of the author in the study and value is the normalized bias value estimated for the study.

4.3.3 Ranking bias score

A total of 48 AI studies are analyzed to identify the RoB in healthcare. For this, we have considered 44 crucial AI-attributes for each study, thus in total 2112 qualitative attributes are generated by assigning a value between 0 and 5 to each attributes. The values are assigned based on the expertise of the AI-scientist's experience in the domain. For

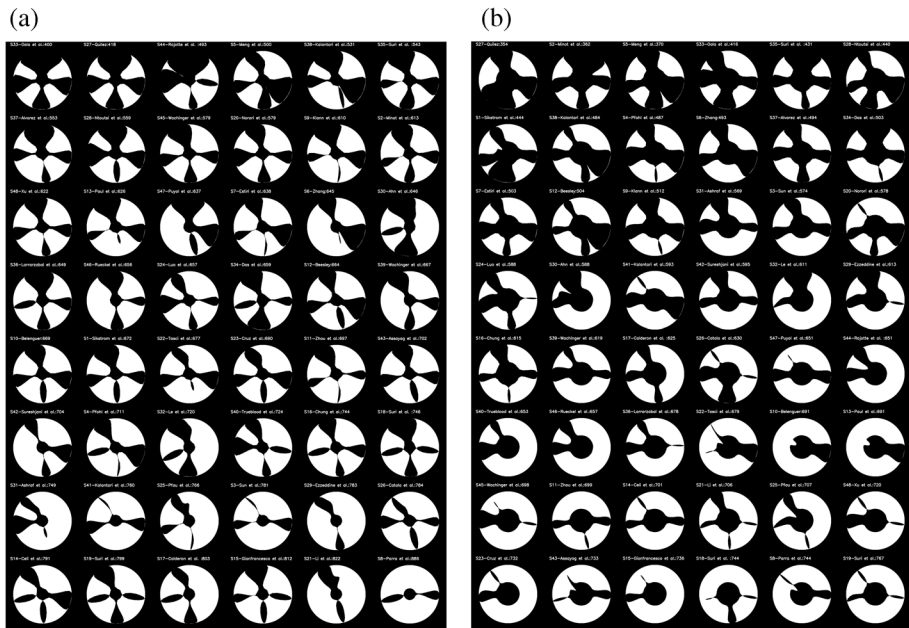


Fig. 6 Bias Configuration Map of 48 selected studies under various classes by regional bias area method for (a) Expert 1 (b) Expert 2

each study, the mean score is calculated by summation of values for all the attributes of the study.

Ranking-bias score method is adopted to evaluate bias in AI-studies in healthcare. The scores are extracted by two AI-scientists. The higher the mean value, the lower is the RoB. Hence, based on the mean score studies are ordered from low-bias to high-bias ranging from 3.5 to 1.1 (from left to right) for expert 1 and 2.9 to 1.0 (from left to right) for expert 2, respectively in decreasing order of their mean score. Based on the mean score value, the studies are organized in three bins namely, *low-bias*, *moderate-bias* and *high-bias* for both the experts as depicted in Figs. 7 and 8, respectively. On the basis of intersection of frequency plot curve and mean score cumulative plot of the studies, the raw cut-off of 2.1 and 1.9 is determined to select the AI-studies for RoB by both the experts, respectively. For expert 1, a total of 22 studies below this cut-off region (ranging from 2.1 to 1.1) are considered in high RoB. Another, 16 studies are in *moderate-bias* region (ranging from 2.5 to 2.1), from left to right in decreasing order. The remaining studies are under *low-bias* region (ranging from 3.5 to 2.5) and have the highest mean value for the attributes scores. Expert 2 categorized 26 studies in *high-bias* (ranging from 1.9 to 1.0), 13 studies in moderate RoB (ranging from 2.4 to 1.9) and 9 studies under *low-bias* region (ranging from 2.9 to 2.4), having the highest mean value for the attributes scores. Studies under low and moderate region have strong AI-design and can be considered suitable for clinical practices in comparison to studies under *high-bias* having low normalized mean score of the attributes.

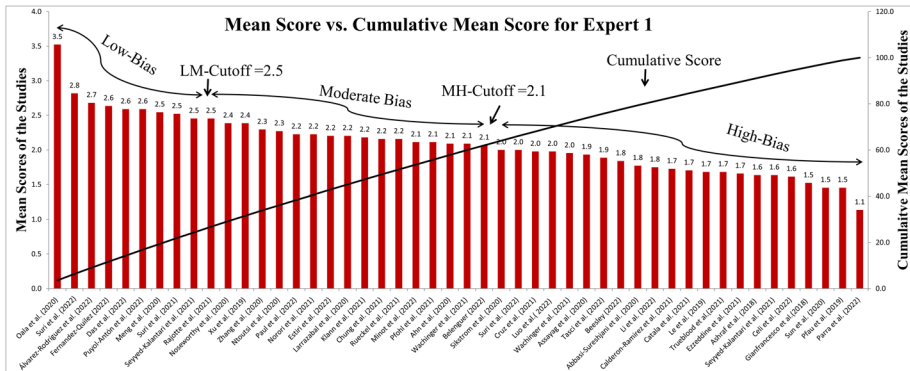


Fig. 7 Frequency plot showing studies in three zones namely, *low-bias*, *moderate-bias*, and *high-bias* with decreasing bias cut-off for Expert 1

4.3.4 Analytical strategy to evaluate AI-bias

In this section, we will discuss about the analytical strategy (ANAL) to evaluate bias in AI studies. For this, the crucial 44 AI attributes are divided into four clusters namely, demographics, architecture, data, and clinical evaluation. The inter-relationship between the attributes in each cluster is determined in accordance with their scores in each study to compute the impact of bias. This will analyze the impact of attributes relationship in a study. The analytical model to compute bias (β) in a study (S) is given in Eq. (1).

$$\beta(S) = 1 - \frac{\sum_{i=1}^C \left(\sum_{j=1}^{n_c} \alpha_{ij}(S) \right)}{\gamma \left(\sum \left((\alpha_1, (\alpha_1 + \alpha_2), \dots, (\alpha_1 + \alpha_2 + \dots \alpha_N)) \right) \right)}, n_c(i) \in N \quad (1)$$

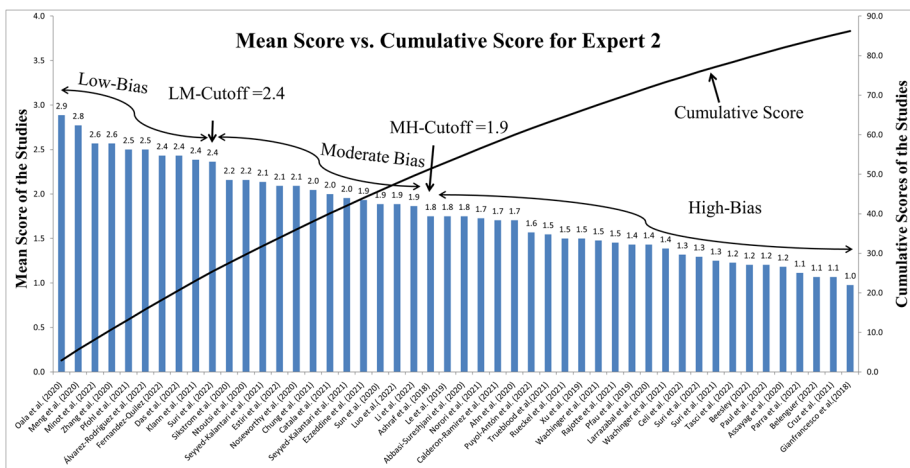


Fig. 8 Frequency plot showing studies in three zones namely, *low-bias*, *moderate-bias* and *high-bias* with decreasing bias cut-off for Expert 2

where, n_c represents the number of attributes α in a cluster C , γ is a parameter regularizer and N is the total number of attributes in a study (S). The numerator of the Eq. (1) will capture the interrelation between the attributes in a cluster. The relationship between the attributes in a cluster is imperative for the microscopically estimating RoB in a study. The denominator will compute the total impact of the attributes score in the bias. This will normalize the overall bias estimation in a study. The function of the parameter regularizer is to maintain the attribute scores between 0 and 1. The overall ratio will determine the bias score of the AI-study. In order to analyze results, the obtained bias estimation values are categorized into three RoB zones namely, *low*, *moderate* and *high*. The number of studies that fall under *low-bias* zone for Expert 1 are **9** [8, 28, 41, 78, 99, 112, 144, 159, 162]. Similarly, studies under *moderate-* and *high-bias* zones are, **12** [15, 34, 40, 62, 92, 97, 109, 111, 119, 131, 141, 143] and **27** [1, 6, 10, 11, 14, 20, 23, 26, 43, 49, 73, 74, 81, 88, 96, 98, 108, 110, 115, 123, 125, 127, 139, 148, 151, 153, 154]. However, Expert 2 categorizes the studies under the three zones as **11** [14, 15, 40, 43, 49, 96, 98, 123, 131, 142, 148], **11** [23, 26, 81, 88, 92, 97, 108, 109, 126, 153, 154], and **26** [1, 6, 8, 10, 11, 20, 28, 34, 41, 62, 73, 74, 78, 99, 110–112, 115, 119, 127, 139, 141, 143, 151, 159, 162]. Analytical results infer that the most of the studies fall under *high-bias* zone. It may be due to that the studies have not considered the imperative evaluation parameters for bias analysis. The bias qualitative analysis is also not performed for the studies to identify the RoB. In addition, the bias mitigation steps such as feature selection and feature augmentation are rarely performed in AI-studies.

4.3.5 Comparison for bias estimation approaches using the Venn diagram

This section will analyze the results of bias estimation techniques namely, RBA, RBM, RBS and ANAL using the Venn diagram (VD). The relationship between the bias estimation techniques for the RoB in three zones such as *low*, *moderate* and *high* for Expert 1 and Expert 2 are presented as VD in Fig. 9a, b & c, respectively. The number of studies categorized in the *low-bias* zone for RBM, RBA, RBS and ANAL by Expert 1 are **10** (20.8%) [8, 43, 88, 96, 98, 99, 115, 126, 142, 154], **10** (20.8%) [8, 43, 88, 96, 98, 99, 115, 126, 142, 154], **10** (20.8%) [8, 34, 43, 88, 99, 112, 115, 127, 141, 142] and **9** (18.7%) [8, 28, 41, 78, 99, 112, 144, 159, 162], respectively. Similarly, number of studies under *moderate-bias* zone for RBM, RBA, RBS and ANAL are **12** (25%) [6, 14, 15, 34, 40, 62, 73, 92, 109, 112, 131, 159], **15** (31.2%) [6, 14, 15, 34, 40, 62, 73, 81, 92, 109, 112, 119, 154, 159, 162], **16** (33.3%) [6, 15, 28, 40, 62, 73, 92, 96–98, 109, 111, 119, 154, 159, 162], and **12** (25%) [15, 34, 40, 62, 92, 97, 109, 111, 119, 131, 141, 143], respectively. Lastly, the number of studies categorized under *high-bias* zone for RBM, RBA, RBS and ANAL are **26** (54.1%) [1, 10, 11, 20, 23, 26, 28, 41, 49, 74, 78, 81, 108, 110, 111, 119, 123, 127, 139, 141, 143, 148, 151, 153, 162, 164], **23** (47.9%) [1, 10, 11, 20, 23, 26, 28, 41, 49, 74, 78, 108, 110, 111, 123, 127, 131, 139, 141, 143, 148, 151, 164], **22** (45.8%) [1, 10, 11, 14, 20, 23, 26, 41, 49, 74, 78, 81, 108, 110, 123, 127, 131, 139, 143, 148, 151, 153], and **27** (56.2%) [1, 6, 10, 11, 14, 20, 23, 26, 43, 49, 73, 74, 81, 88, 96, 98, 108, 110, 115, 123, 125, 127, 139, 148, 151, 153, 154].

However, Expert 2 categorize studies under *low-bias* for RBM, RBA, RBS, and ANAL as **9** (18.7%) [34, 43, 88, 92, 98, 99, 111, 131, 142], **11** (22.9%) [8, 43, 88, 92, 98, 99, 111, 126, 131, 142, 162], **10** (20.8%) [8, 34, 43, 62, 88, 92, 99, 111, 142, 162], and **11** (22.9%) [14, 15, 40, 43, 49, 96, 98, 123, 131, 142, 148], respectively. For *moderate-bias*, expert 2 categorizes studies for RBM, RBA, RBS and ANAL as **12** (25%) [8, 10, 14, 23, 40, 62, 81, 96, 126, 127, 139, 163], **11** (22.9%) [1, 6, 10, 14, 34, 40, 62, 81, 96, 127, 139], **12** (25%)

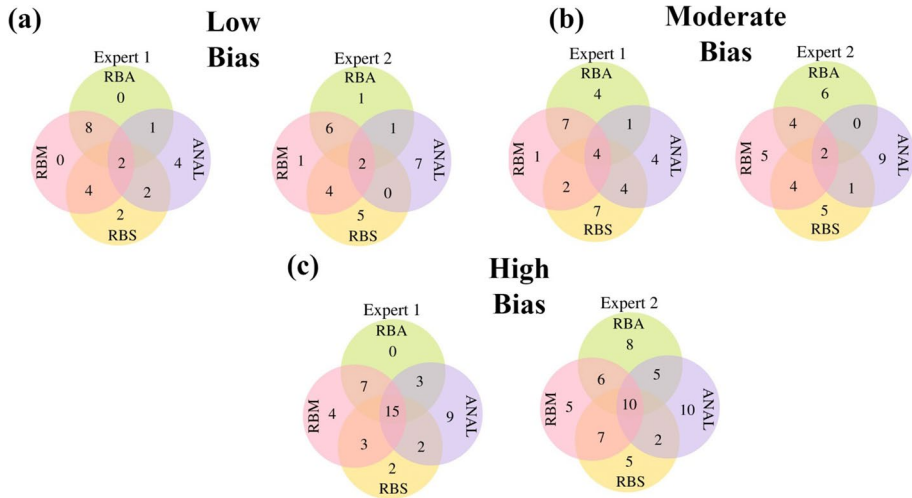


Fig. 9 Comparison of four bias estimation methods namely, RBM, RBA, RBS and ANAL using Venn diagram for Expert 1 and Expert 2 for the three zones (a) Low-bias (b) Moderate-bias (c) High-bias

[23, 28, 40, 41, 78, 81, 97, 98, 126, 127, 131, 139] and **11** (22.9%) [23, 26, 81, 88, 92, 97, 108, 109, 126, 153, 154]. Finally, the number of studies categorized under *high-bias* zone for RBM, RBA, RBS and ANAL are **27** (56.2%) [1, 6, 11, 15, 20, 26, 28, 41, 49, 73, 74, 78, 108–110, 112, 115, 119, 123, 141, 143, 148, 151, 153, 154, 159, 164], **26** (54.1%) [11, 15, 20, 23, 26, 28, 41, 49, 73, 74, 78, 108–110, 112, 115, 119, 123, 141, 143, 148, 151, 153, 154, 159, 164], **26** (54.1%) [1, 6, 10, 11, 14, 15, 20, 26, 49, 73, 78, 96, 108–110, 112, 115, 119, 123, 141, 143, 148, 151, 153, 154, 159] and **26** (54.1%) [1, 6, 8, 10, 11, 20, 28, 34, 41, 62, 73, 74, 78, 99, 110–112, 115, 119, 127, 139, 141, 143, 151, 159, 162]. The number of studies lie in the overlap region of *low-bias*, *moderate-bias* and *high-bias* zones for four estimation methods for Expert 1 are **2** [99, 142], **4** [15, 40, 62, 109], and **15** [1, 10, 11, 20, 23, 26, 49, 74, 108, 110, 123, 127, 139, 148, 151] and for Expert 2 are **2** [43, 142], **2** [81, 126] and **10** [11, 20, 73, 110, 112, 115, 119, 141, 143, 151] respectively.

4.3.6 Inter-variability analysis

The intra-variability observations for the bias estimation between both the experts, it has been observed that the studies ([142]) and ([20, 110, 151]) are categorized in the *low-bias* and *high-bias* by both the experts in all the four bias estimation strategies, respectively. This overlap has shown significant agreement for the intra-observer variability. Also, intra-variability observations between the mean score of various studies computed by the two AI experts are depicted in Fig. 10. In this, there is a strong overlap between the outcomes of both the expert where studies such as ([8, 34, 43, 88, 99, 142]), ([28, 40, 94, 97, 98]) and ([1, 10, 11, 14, 20, 26, 49, 78, 108, 110, 123, 143, 148, 153]) are categorized by both the expert in three zones namely, *low-bias*, *moderate-bias* and *high-bias*, respectively. Conversely, there is also minor disagreement between the results of both experts. Studies such as ([62] & [111]) are categorized in *low-bias* region by expert 2 but expert 1 evaluated them in *moderate-bias* region. In addition, there is also a steep difference between the bias

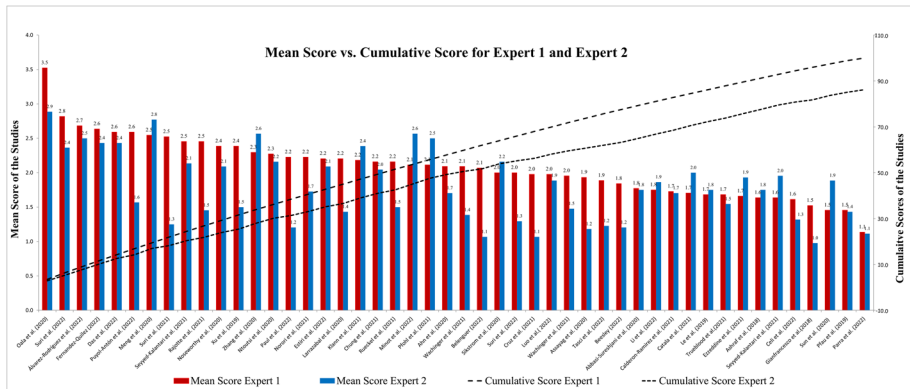


Fig. 10 Mean Score vs. Cumulative Score plot for Expert 1 and Expert 2

mean score computed for the studies [112, 115], and [141] by both experts. It may be due to the reason that expert 2 not able to analyze the crucial AI-attributes from the mentioned studies. Also, the expert 2 domain knowledge in AI cannot able to extract the hidden AI-attributes due to less transparency and elaboration by the authors in their studies.

The correlation between the results of AI experts is determined by statistical measures namely, t-test and Spearman's rank coefficient. The estimated p-values are **0.0033** and **0.0010** for the studies using t-test and Spearman's rank coefficient, respectively. Similarly, the p-value analysis for the selected AI attributes for each statistical test is obtained as **0.0834** and **2.9E-09**. The results infer that the studies and AI attributes are strongly correlated as the computed p-value is statistically significant. In addition, the attributes scores for the crucial 44 AI attributes are illustrated in Fig. 11. It has been observed that both experts have almost related mean score values for AI attributes namely, age, data type, synthetic data, kinship, precision, recall, and p-value. However, the mean values for the AI attributes such as data quality, risk analysis, unseen data, robustness and qualitative bias test are relatively high for expert 2 compared to expert 1. It is due to the insufficient transparency and explanations by the authors in the paper which is captured by the expert 2. The expert 2 may have overrated the AI attributes in particular studies due to which high scores are deliberated.

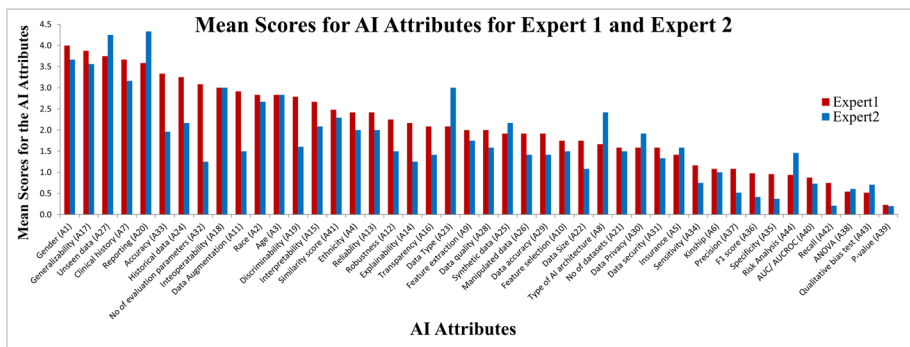


Fig. 11 AI Attributes Mean Score for Expert 1 and Expert 2

5 Bias mitigation approaches and legal manifestations

Mitigating bias is imperative for the fairness and clinical deployment of the AI-model in practice. For this, bias should be identified at all three steps of AI-model namely, pre-processing, in-processing and post-processing. Mitigation efforts at pre-processing step primarily analyze the process to remove data bias [103]. In-processing step deals with factors to mitigate algorithmic bias [16]. Bias due to AI-model is focused and mitigation efforts are determined at the post-processing step for discrimination-aware classification [31].

5.1 Bias mitigation

Mitigating bias at pre-processing step emphasizes the requirement of bias due to data collection. Bias during data collection can be avoided by incorporating diversity in the data from various subgroups [140]. Information about the data collection source should be made public so that certain assumptions can be considered at the model development level. Characteristics, motivations, limitations, and assumptions in the dataset should be appropriately documented [17]. In addition, datasheet for the dataset should be prepared to highlight the transparency in the data collection process and specify the potential risk (if any) in the dataset usage [46]. Historical and societal bias in the datasets can be avoided to generate a balanced dataset for generalizability of the model. Social and ethical barriers in relation to big data can be evaluated for differential diagnosis [86]. Calmon et al. [21] proposed a mapping approach to map the original data points with the newly created datasets for incorporating demographic information. The method aimed to generate a balanced dataset including protected and unprotected attributes information. Techniques to mitigate bias at in-processing step proposed to modify the model learning process. To improve the model discrimination during learning process, it is better to suppress the sensitive features, change the objective function or alter the AI design parameters such as loss function, regularizer, and optimization function. In [60], authors proposed integrating a regularizer in the model to reduce the correlation between the sensitive features and input class labels. Nagare et al. [94] proposed a loss function to minimize bias in training weights. Also, iterative training was recommended to address the change in in-training weights for the input classes of the model [63]. Cumulative fairness techniques and constraint-based approaches were used to treat the class imbalance error in the model [160] [54]. Adversarial learning along with variable of group learning was presented to mitigate undesired bias from the well trained model [161].

Strategies to mitigate bias at post-processing step include adaptive classifier, facts in the decision making and testing on multiple datasets. Classifiers should be formulated constraint-free to ensure fairness in the outcome [158]. Authors proposed adaptive reweighting of the training weight for fairness-aware classification [63]. In another line of work, ranking framework was proposed to mitigate the impact of bias due to classification error [59]. Post-processing techniques ensure that the learned model will be treated as black box for all the training data for generalizability and interoperability of the model. This stage also deals with the explainability and interpretability of the outcome due to complex model design. In sum, pre-processing steps, highlights the requirement for balanced data to mitigate bias in the dataset. The feature engineering and model design parameters are checked

to mitigate bias at the in-processing step. Finally, the model is checked for non-discriminative outcomes at the post-processing step.

5.2 Legal manifestations

Data modification and model alteration techniques to mitigate bias at pre-, in- and post-processing step of AI-model attract legally pertinent questions. Intellectual property rights encourage defining legal provision laws for data collection, alteration, processing, and modification. Besides this, data ownership and data privacy are two major concerns that need to be addressed before data usage either for research purposes or for commercial benefits in healthcare. Ethical, legal, and social perspectives are imperative to understand for defining the rights related to data collection, data sharing, and data privacy. Legal laws for data protection are country-specific which deal with personal data usage, re-usage, and modification [47]. For instance, data regulatory body such as General Data Protection Regulation (GDPR) was formulated in Europe to safeguard patient's personal data sharing and modification. The law has given patients full authority to either provide or withdraw consent at any time for their medical data usage and sharing [53]. However, this step toward data security has restricted the model development. The concern is what will happen if the patient withdraws the consent just before the final model deployment. Then, the developer has the choice to either discard the complete model parameters or re-develop the system with new training parameters. Hence, data ownership and protection laws should be vulnerable to these modifications in the interest of prediction model development in healthcare. Legally binding regulations must be flexible enough to accommodate the suggestions and recommendations from independent bodies for effective, efficient and transparent model development in healthcare designs.

The increased demand for AI-model training data raises concerns over data quality, accuracy, and truthfulness. Data availability in healthcare training is limited due to the limitations and challenges in clinical data collection, usage, and patient privacy issues [114]. It is advisable to generate synthetic data from the original data to increase the training data for the AI-model [91]. Synthetic data not only increases the amount of training data but also minimizes the risk associated with patient data sharing and privacy. In addition, synthetic data also addresses the bias in training data due to unbalance datasets. However, if the original data suffers from bias then, synthetic data will intensify it in the training dataset and make the prediction model unstable. Further, plenty of open sharing dataset repositories are available in public domain for AI model training. Most of the public dataset repositories are not diverse, interoperable, and generalizable in their applications [50]. In addition, these datasets are inconsistent in their data collection, aggregation, and truthfulness process. Utilizing these datasets in complex and sensitive systems such as healthcare is not recommended. Hence, evaluating the quality and accuracy of the datasets utilized in AI-models in healthcare is advisable. Quality standards must be impactful to generate high-quality, consistent, and interoperable datasets for the clinical deployment of the model.

Data protection, sharing and privacy law cover both the input as well as the output of automated AI-model decision-making. Unawareness about the sensitive data attributes and potential bias in input dataset will lead to non-discriminatory outcomes by AI-model towards a subgroups. Discrimination in datasets questions the fairness of the AI-model. Discrimination in healthcare designs lead to underdiagnosis or misdiagnosis

to underrepresented subgroups of the population. It restricts the generalizability and applicability of the algorithm to all the individuals belonging to a particular group [61]. Anti-discrimination rules to define legal policies that may prevent dissemination of discrimination against certain group in the algorithm [163]. Anti-discrimination directives should be broad enough to accommodate all unlawful criteria for automated decision-making of AI-model [155].

To summarize, data sharing in healthcare has raised concerns over patient's privacy and security. The data security norms and regulations have restricted the amount of data availability to AI-model. To meet the data-hungry demand of AI-model, it is suggested to generate synthetic data. But the clinical applicability of the model trained on synthetic data needs to be evaluated. In addition, data gaps in publicly available data repositories need to be addressed to ensure the data quality and accuracy in healthcare. Anti-discrimination laws need to be robust enough for equal treatment to all individuals of the population.

6 Discussion

In this study, based on the input data the AI-studies are categorized into three clusters namely, PD-ai, ID-ai, and HD-ai in the healthcare domain. A total of 44 crucial AI attributes were identified and the mean score for the studies were calculated. Four bias estimation techniques RBM, RBA, RBS, and ANAL were exploited to identify the RoB in AI-studies. RBM computes bias in a study by analyzing the link between the low-performing attributes and high-performing attributes. RBA is a method based on regional area for bias computation in a study. On the other hand, RBS ranks the AI-studies based on the mean scores of the attributes. ANAL methods validate the results obtained by the other methods. It has been observed that two cutoffs **2.5** and **2.1** yields the **27%**, **27%**, **21%** and **22.9%** in *low-bias* region for RBM, RBA, RBS and ANAL, respectively, by expert 1.

6.1 Benchmarking table

Benchmarking Table 5 represents the comparison of the proposed work with seven other existing studies, considering eight crucial attributes. The last row of the table consists

Table 5 Benchmarking Table

SN	Citation	3-BCAI ^a	PRISMA ^b	APP ^c	#STD ^d	DA ^e	BC ^f	BR ^g	BAE ^h
1	Parikh et al. (2019) [107]	×	×	Generic	8	✓	×	×	×
2	Noseworthy et al. (2020) [97]	×	×	Specific	18	✓	×	×	×
3	Gurupur and Wan (2020) [52]	×	×	Generic	22	×	×	×	×
4	Byrne (2021) [19]	×	×	Generic	19	✓	×	×	×
5	Sousa et al. (2021) [104]	×	×	Specific	27	×	×	×	×
6	Suri et al. (2022) [142]	×	✓	Specific	70	×	✓	✓	×
7	Gallifant et al. (2022) [45]	×	✓	Specific	95	✓	✓	×	×
8	Kumar et al. (proposed)	✓	✓	Generic	72	✓	✓	✓	✓

[3-BCAI^a: 3- Bias classes of AI-Model, APP^c: Type of Medical Application, #STD^d:# Number of Studies, DA^e: Demographic Analysis, BC^f: Bias Computation, BR^g: Bias Ranking, BAE^h: Bias Analytical Evaluation]

of the proposed study. To represent the presence and absence of the unique attribute in a particular study, we have marked “✓” and “✗”, respectively. It has been observed that none of the work has categorized the bias into three AI-classes. Most of the works analyze bias either to the specific or generic medical fields. Applications to specific medical domains include mechanical ventilation [45], COVID CT-Scans [104], ECG [97] and COVID X-rays [144], whereas generic refers to assessing bias at different stages of AI-model in medical domain [19, 52, 107]. Very few studies discuss about the impact of bias and its computation [45, 142]. The bias ranking is also explored by very limited researchers [144]. The strength of the proposed study is the analysis of AI-bias by categorizing the medical AI-models into three categories. In addition, bias interpretation and bias computation based on 44 crucial and unique AI attributes is the main potential of the work to assess the possibility of practical implications of AI-model in the medical domain.

6.2 Recommendations/challenges

Based on the AI-bias study and AI-attributes statistical analysis, recommendations are being clustered for the upcoming AI-studies. The recommendations are (a) Inclusion of sociodemographic information (b) AI-architectural vision (c) Dataset design and specification and, (d) Clinical evaluation and validation. The discussion of these recommendations is as follows: (i) Inclusion of sociodemographic information: The studies utilize gender, age, racial and ethnicity to identify the bias in the AI-studies. Most of the studies (27%) have zero or one sociodemographics data. The majority of the studies have neglected kinship (more than 95%) and insurance (90%) attributes which are essential to identify socioeconomic status of the patient to prevent the underdiagnosis [127]. (ii) AI-architectural vision: In Medicine, the AI-architectural design must be explainable and interpretable to understand the sensitive feature contribution towards outcome. Sometimes the prediction associated with explainable features leads to adverse impact on clinical outcomes. In addition, the black box nature of AI-model architecture restricts the generalizability and interoperability of the model which may cause incorrect diagnosis in the prediction [55, 104]. (iii) Dataset design: The dataset used for the evaluation should be balanced enough in terms of sociodemographic features for addressing the societal bias in the outcome. The documentation can be provided mentioning the associated risks and limitations for utilization of the dataset. The AI-model should also specify the test and training specifications so that the impactful scenarios can be determined. However, none of the present studies specify the structural details of the dataset explicitly. The real-time validation of the AI-model in healthcare is not possible to prevent harmful outcomes. (iv) Clinical evaluation and validation: There is no unique distribution of evaluation metrics in the AI-model to analyze and compare the performance. All the AI-attributes are equally important and hence, require common performance metrics to compare the outcomes. The AI-model must also be evaluated on unseen data to check the accuracy in the prediction. In addition, it is also mandatory to clinically validate the outcomes of AI-model for the implementation of the studies from paper to practice.

6.3 The future of bias in healthcare system designs: a short note

While the study of AI bias in healthcare systems is complete in itself, the field of AI is evolving at a rapid pace and new advances have stepped which affect the bias in several ways. There are three rapidly burning areas of AI that need to be seen as to how this will shape AI. (a) This includes the role of pruning techniques in AI paradigm to reduce storage and improve the speed considerations [3, 4]. Although, this topic is out of the scope of the current work, we see in the future the role of pruning affecting the AI-bias and bias in healthcare system design. (b) The second aspect of the AI bias is the evolving algorithms are based on the fusion of individual techniques such as ensemble paradigms. Different classification methods are fused for superior results reducing bias by handling the inconsistencies and variability in the data sets [58, 144, 147]. This is yet to be analyzed and seen in the future. (c) The last and third important aspect of AI is the role of big data framework where the large number of sources of data can be fused into larger cohorts [12, 38].

7 Conclusion

Our study focuses on identifying the source of bias in AI-model. To study the bias, 72 key studies were selected using PRISMA model. Out of these studies, 48 studies are categorized under three classes such as point data-based AI-model (PD-ai), image data-based AI-model (ID-ai), and hybrid data-based AI-model (HD-ai). Data bias and algorithmic bias in design of AI-based medical systems in these three categorizes is identified using 44 AI-attributes. To analyze the risk of bias in a study, the strength of these AI attributes is evaluated using four efficient bias computation models namely, RBA, RBM, RBS and ANAL. Our study have analyzed that studies that lack in accommodating the crucial AI-attributes are more prone to RoB. Our study have also revealed that the majority of the studies in the healthcare suffered from data bias and algorithmic bias. It may be due to the black box nature of the algorithmic design and incomplete specification of the training data. In addition, legal issues related to data sharing, privacy and security are also responsible to exacerbating bias in AI-models. It is recommended that before the practical implication of AI-model in medical healthcare system the robust and effective analysis should be performed with appropriate clinical validation and approval.

Appendix

Table 6 Bias attributes details of the representative work under PD-ai model

SN	Citation	Bias Type	Data Source	Bias Accountability	Bias Computation	Summary
1	Sikstrom et al. (2020) [131]	Data bias, sample bias, understanding bias, implementation bias	Data from 213 published literature in various journals	Explainability, interpretability, transparency, impartiality, provenance	Analyzed to explore issues and provide solution for inaccurate, underdiagnosis prediction in psychiatric, violence, aggression prediction in dementia patients	Provided a conceptual multidimensional framework with three pillars of fairness to identify issues and bias in psychiatric related health issues for multiple stakeholders
2	Minot et al. (2022) [92]	Gender bias, dataset bias	n2c2 deidentification challenge and medical information mart for intensive care-III database	Interpretability	Matthew's correlation coefficient (MCC), cosine similarity scores, intrinsic bias within word-embedding spaces, empirical bias in word-frequency distributions, rank turbulence divergence	Proposed a de-biasing methodology based on BERT-gender classifiers for identification and removal of gender bias from clinical notes
3	Sun et al. (2020) [139]	Gender bias	Data from Columbia University Irving Medical Center, MIMIC-III	Transparency, discriminability	Recall, mean squared error (MSE), mean squared discrimination	Identified gender disparities in time to diagnosis based on the medical history and patient's data
4	Pföhl et al. (2021) [111]	Sociodemographic bias, societal bias	Stanford Medicine Research Data Repository, Optum's de-identified clinical data mart database, MIMIC-III	Discriminability	AUROC, relative calibration error, average precision, mean differences, prediction parity, relative calibration and ranking	Proposed to penalize differences between the distributions of predictions across groups induce nearly universal degradation of multiple performance metrics within groups for conceptualizing the algorithmic fairness

Table 6 (continued)

SN	Citation	Bias Type	Data Source	Bias Accountability	Bias Computation	Summary
5	Meng et al. (2020) [89]	Sociodemographic bias, societal bias, algorithmic bias	MIMIC-IV	Interpretability, reliability, transparency	AUPRC, AUROC, Jaccard similarity,	Explained interpretability of the model based on the feature importance ranking based on the sociodemographic information available in the dataset
6	Zhang et al. (2020) [162]	Gender bias and racial bias	Varying types of clinical notes	Diversity in data	Precision, recall, parity gap, AUPRC, AUROC, SPE	Demonstrate quantitatively and qualitatively word embedding that propagate unwanted latent relationships with regards to different genders, language speakers, ethnicities, and insurance groups
7	Estiri et al. (2022) [40]	Algorithmic bias, sociodemographic bias, training bias	RT-PCR medical data	Reliability, interpretability, discriminability	AUROC, Brier Score, Wilcoxon rank-sum test, p-value correction, and MAE	Predicted severe acute respiratory syndrome coronavirus 2 severity risk based on the medical data
8	Parra et al. (2022) [108]	Data bias, racial bias, gender bias	U.S. based individual participants	Data training awareness and discrimination	Average likelihood for questioning-based AI recommendation systems, p-value	Determining bias in seven scenarios including healthcare in questioning-based AI recommendation systems
9	Klann et al. (2021) [62]	Data bias	Data from 12CE clinical sites	Robustness, generalizability, interoperability	AUROC, SEN, SPE, positive predicted value, negative predicted value, F1 score and confidence index (CI)	Covid-19 severity analysis by adopting electronic health record data from 12 international clinical sites
10	Belenguer (2022) [15]	Individual bias, discriminatory bias, societal bias	-	Explainability, interpretability, transparency, accountability, discriminability	Empirical and technical value evaluation, COMPAS, Blind testing	Discussed the discriminatory bias in the AI based solution adapted by the pharmaceutical industry

Table 6 (continued)

SN	Citation	Bias Type	Data Source	Bias Accountability	Bias Computation	Summary
11	Beesley (2022) [14]	Selection bias, data bias	Michigan Genomics Initiative repository	Robustness, interpretability, generalizability	Proposed weighting methods to correct selection bias in EHR data	Proposed weighting methods to correct selection bias in EHR data
12	Noseworthy et al. (2020) [97]	Data bias, racial bias, ethnicity bias	Patient's ECG data	Interpretability, accountability, generalizability	AUC, ROC, SPE, SEN, ACC	Utilized ECG features to detect left ventricular ejection fraction (LVEF) and concluded that LVEF is invariant to race
13	Paul et al. (2022) [109]	Data bias, algorithmic bias	Parkinson's disease (PD) related research literatures	Interpretability, accountability	ACC, SPE, SEN, AUC, MCC, negative predictor value, Dice coefficient	Analyzed risk of bias in AI based PD studies and categorized them into three bins to rank them from high risk to low risk for clinical validation
14	Celi et al. (2022) [26]	Data bias, algorithmic bias, gender bias, ethnicity bias	Published AI based clinical studies in PubMed in 2019	Reliability	Confusion matrix, ROC	Demonstrated that most of the clinical work primarily presented by male authors without clinical knowledge
15	Gianfrancesco et al.(2018) [49]	Sampling bias, algorithmic bias	-	Transparency, reliability, discrimination	Suggested recommendation in clinical decision system to identify bias due to missing sociodemographic and socio-economic information in the healthcare records	Analyzed healthcare disparities in healthcare record due to missing data leading to misclassification error
16	Chung et al. (2021) [28]	Gender bias, algorithmic bias, training bias	Patient's clinical information	Discrimination	True positive, false negative, false positive, true negative, SEN, SPE, ACC, balanced ACC, AUC	Investigated that model prediction can be biased if model was trained on particular type of datasets. Used clinical information to predict the severity of COVID-19

Table 7 Bias attributes details of the representative work under ID-ai model

SN	Citation	Bias Type	Data Source	Bias Accountability	Bias Computation	Summary
1	Calderon-Ramirez et al. (2021) [20]	Data bias, algorithm bias	Patient's Chest X-ray images	Reliability, generalizability	ACC, F1-score, Confidence measure, JS divergence for similarity estimation	Evaluated three uncertainty estimation methods for accurate prediction of covid-19
2	Suri et al. (2021) [141]	Data bias, selection bias, measurement bias, bias due to missing data	AI studies based on CXR and CT imaging	Risk granularity, comorbidity, reproducibility, data quality, reliability, robustness	ACC, SPE, SEN, F1-score, precision, recall, ROBINS-I, PROBAST	Qualitatively and quantitatively analyzed the risk of bias in AI studies and rank them on the basis of 10 AI attributes
3	Suri et al. (2022) [142]	Sampling bias, data bias, algorithmic bias	AI studies based on CXR and CT imaging	Risk granularity, accountability, reliability, robustness	ROBINS-I, PROBAST, RBM, RBA, RBS, Venn diagram	Five strategies were proposed to analyze bias and categorized studies in three bins i.e., low bias, moderate bias and high bias
4	Norori et al. (2021) [96]	Data bias, algorithmic bias, human bias, social bias	Examples of skin lesion images in healthcare	Explainability, interpretability, reliability, interoperability, inclusivity, data standards, transparency	F1-score	Analyzed data gaps, data privacy, ethical issues for unbiased big data in healthcare
5	Li et al. (2022) [78]	Data bias, source bias, training bias	CT scans data from seven hospitals	Generalizability, interoperability	SEN, SPE, F1-score, ACC, AUC, processing time, t-test validation, p-value	Proposed a simple and efficient methodology for dealing with training bias in the datasets exploited for covid-19 prediction
6	Tasci et al. (2022) [148]	Data bias, algorithmic bias, training bias	Oncology image dataset	Discrimination, generalizability, interpretability	SEN, SPE, F1-score, ACC, AUC, MCC	Identified the potential sources of bias in the datasets and summarize the methods to handle unbalanced bias in oncology datasets

Table 7 (continued)

SN	Citation	Bias Type	Data Source	Bias Accountability	Bias Computation	Summary
7	Cruz et al. (2021) [123]	Dataset bias, selection bias	Chest CXR imaging datasets	Generalizability, interpretability, transparency, dataset quality standards	CHARMS, bias tools	Identified risk of bias in publicly available medical imaging datasets
8	Luo et al. (2022) [81]	Dataset bias	Self-generated CXR datasets	Generalizability, robustness, interpretability,	AUC	Proposed pseudo bias balanced learning algorithm for debiasing no explicit bias attribute labeling and generated biased datasets from public datasets
9	Pfau et al. (2019) [110]	Dataset bias, training bias, model bias	Skin lesion imaging dataset	Interpretability, inter model operability	p-value, global saliency, similarity occurrence between class and artifact	Evaluated the model bias induced by the model artifacts using global saliency framework
10	Catala et al. (2021) [23]	Dataset bias	CXR image datasets	Generalizability, interpretability, discrimination	Accuracy, AUC ROC,	Analyzed bias in medical imaging classification by combining datasets from different sources
11	Fernandez-Quilez (2022) [43]	Data bias, sampling bias, selection bias, social bias	MRI images	Explainability, interpretability, transparency, accessibility, generalizability	Bias in data collection, data ownership	Discussed about the quality, ownership, privacy and rights of the medical imaging data used by AI-models
12	Ntoutsis et al. (2020) [98]	Data bias, socio technical bias	-	Explainability, interpretability, discrimination, data accuracy	-	Discussed methods for understanding, mitigating and accounting bias, along with associated legal aspects

Table 7 (continued)

SN	Citation	Bias Type	Data Source	Bias Accountability	Bias Computation	Summary
13	Ezzeddine et al. (2021) [41]	Data bias	CXR and CT imaging data	Explainability, usability	Precision, recall, ACC, F1-score	Analyzed that model had good accuracy when trained and tested on same dataset in comparison to different datasets
14	Ahn et al. (2020) [6]	Algorithmic bias	Three medical imaging datasets	Robustness, transparency	ACC, AUC	Developed a DL based model for classification of medical data based on context feature augmentation
15	Ashraf et al. (2018) [10]	Dataset bias	Four brain MRI datasets	Transparency, interoperability, discriminability	ROC AUC, ACC	Presented a method a unlearn data membership to handle data bias in MRI datasets
16	Le et al. (2019) [74]	Dataset bias, automation bias	Breast ultrasound, breast tomosynthesis, breast MRI	Explainability, interpretability, data quality, data privacy	Highlighted the usage of AI-CAD tools in medical imaging	Highlighted the importance of analyzing embedded bias in CAD tools before the implication

Table 8 Bias attributes details of the representative work under HD-ai model

SN	Citation	Bias Visualization	Data Source	Bias Accountability	Bias Computation	Summary
1	Oala et al. (2020) [99]	Gender bias, age bias, society bias	Retina imaging, brain MRI scans, single cell images	Robustness, explainability, interpretability, reporting	ACC, precision, false negative rate, false omission rate, SEN, SPE, F1 score, AUCROC, negative predicted value, precision and predicted prevalence, positive predicted value, CI, aequitas analysis	Analyzed three medical use cases and highlighted the importance of bias evaluation in medical prediction models. Also, emphasized the requirement of WHO healthcare audit framework for quality assessment of healthcare models
2	Das et al. (2022) [34]	Algorithm bias, segmentation bias, training bias	CT scans, MRI scans	Robustness, explainability	ACC, SEN, SPE, ROC, dice coefficient, RBS	Identified risk of bias in brain image segmentation methods and ranked them in low, moderate and high range based on 32 attributes
3	Suri et al. (2022) [143]	Algorithm bias, data bias, gender bias, age bias, ethnicity bias	carotid ultrasound images, and patients cohorts biomarkers	Discriminability	ACC, precision, false negative rate, false omission rate, sensitivity, specificity, F1 score, AUC, negative predicted value, precision, positive predicted value, confidence interval, ANOVA, Kappa coefficient	Cardiac vascular model based on point data only analyzed under high bias risk in comparison to the models based on combination of point data and image data. Forty six attributes were quantified to analyse the risk of bias in studies
4	Larrazabal et al. (2020) [73]	Gender bias, data bias, training bias	CXR datasets	Discriminability	AUC, p-value	Images were labeled for multiple disease and data was split randomly to mitigate bias
5	Álvarez-Rodríguez et al. (2022) [8]	Gender bias, data bias	CXR datasets	Discriminability, generalizability	Precision, recall, F1-score, ACC	Proposed a system for the identification of covid-19 addressing sex and age imbalance in the datasets

Table 8 (continued)

SN	Citation	Bias Visualization	Data Source	Bias Accountability	Bias Computation	Summary
6	Seyyed-Kalantari et al. (2021) [127]	Gender bias, age bias, data bias, training bias, ethnicity bias human bias, historical bias	CXR datasets, multi-source dataset	Discriminability, generalizability	False negative rate, false positive rate, ROC	Identified risk of unequal treatment on the basis of gender and ethnicity in the AI based healthcare model
7	Wachinger et al. (2021) [154]	Gender bias, age bias, feature bias, confounding bias	MRI neuroimaging datasets	Discriminability	ACC, AUC, Wilcoxon signed-rank, p-value	Demonstrated the induction of bias in CXR datasets and methods to evaluate and reduce the effectiveness of bias
8	Trueblood et al. (2021) [151]	Algorithm bias, evaluation bias	Patient's white blood cell images	Discriminability, accountability	Credible interval, posterior mean, Deviance Information Criterion,	Illustrated that image based decision making can apply accumulation modeling to disentangle different types of biases
9	Seyyed-Kalantari et al. (2021) [126]	Gender bias, race bias, age bias, socioeconomic bias	Three CXR imaging datasets and one multi-source CXR dataset	Generalizability, interoperability	False positive rate, false negative rate	Identified underdiagnosis by AI model for underserved subpopulation and intersectional subgroups
10	Abbasi-Sureshjani et al. (2020) [1]	Gender bias, age bias	Skin lesion imaging dataset	-	AUC	Emphasized that context attributes such as skin color or hair color should be included in the training data for unbiased output
11	Assayag et al. (2020) [11]	Gender bias, age bias, bias due to multivariate information	Patient's clinical information, CT scans, lung function and bronchoalveolar lavage data	Interpretability, accountability, discriminability	CI, p-value, multi-variants logistic regression coefficient	Analyzed the gender bias for the prediction of idiopathic pulmonary fibrosis as male patients are more prone to this than female patients

Table 8 (continued)

SN	Citation	Bias Visualization	Data Source	Bias Accountability	Bias Computation	Summary
12	Rajotte et al. (2021) [115]	Data bias, training bias, location bias	Image datasets and skin lesion datasets	Interpretability, accountability, discriminability, interoperability	ACC, AUC-ROC	Multiple image features exploited for skin lesion classification and generated synthetic data for unbiased results
13	Wachinger et al. (2021) [153]	Gender bias, age bias, data bias, confounding bias	MRI scans from neuroimaging datasets	Generalizability, interoperability	ACC, Mean-difference	Quantify the confounding and casual bias in neuroimaging datasets exploiting casual inference
14	Rueckel et al. (2021) [119]	Confounding bias, training bias	CXR images	Discriminability	AUC, AUCROC, SEN, SPE,	Pneumothorax location, size and shape labeled in CXR images for accurate detection
15	Puyol-Antón et al. (2022) [112]	Data bias, social bias, measurement bias, training bias	Patient's cardiac magnetic resonance images (CMR)	Discrimination, accessibility	Dice score, absolute/relative errors in measurement, ANCOVA analysis, p-value, Dice similarity coefficient, CI	Analyzed racial and gender bias in CMR segmentation model due to imbalance datasets
16	Xu et al. (2019) [159]	Dataset bias	581 CT scans from 178 patients and 178 CT scans from 89 patients	Reliability, generalizability, interoperability	AUC, p-value, CI Wilcoxon rank test, survival test	Considered time series follow up to analyze that young patients recover fast post-surgery and pathological lungs cancer results were independent of clinical information such as gender, tumor size etc

Table 9 Grading Scheme for evaluation of the AI attributes

Clusters	Attributes	Grading Scheme
Socio-demographics (C1)	Gender (A1)	If considered= 5, else= 1
	Race (A2)	If considered= 5, else= 1
	Age (A3)	If considered= 5, else= 1
	Ethnicity (A4)	Multiethnic = 5, Mono ethnic = 2, Not mentioned = 1
	Insurance (A5)	If considered= 5, else= 1
	Kinship (A6)	If considered= 5, else= 1
	Clinical history (A7)	More than 5= 5, 3–2 =2, not mentioned=1
	Type of AI architecture (A8)	DL= 5, ML=3, Not mentioned = 1
	Feature extraction (A9)	If considered= 5, else= 1
	Feature selection (A10)	If considered= 5, else= 1
	Data Augmentation (A11)	If considered= 5, else= 1
	Robustness (A12)	If considered= 5, else= 1
	Reliability (A13)	If considered= 5, else= 1
	Explainability (A14)	If considered= 5, else= 1
	Interpretability (A15)	If considered= 5, else= 1
	Transparency (A16)	If considered= 5, else= 1
	Generalizability (A17)	If considered= 5, else= 1
	Interoperability (A18)	If considered= 5, else= 1
	Discriminability (A19)	If considered= 5, else= 1
	Reporting (A20)	If considered= 5, else= 1
Data Bias (C3)	No of datasets (A21)	Above 3 = 5, from 3=3, 2=2, 1= 1 , not mentioned = 0
	Data Size (A22)	Dataset Size: 5000 and more than =5, 5000–4000=4, 4000–3000=3, 3000–2000=2, 2000–1000=1, less than 1000= 0
	Data Type (A23)	Hybrid data=5, image data=3, point data =1
	Historical data (A24)	If considered= 5, else= 1
	Synthetic data (A25)	If considered= 5, else= 1

Table 9 (continued)

Clusters	Attributes	Grading Scheme
Bias Evaluation (C4)	Manipulated data (A26)	If considered= 5, else= 1
	Unseen data (A27)	If considered= 5, else= 1
	Data quality (A28)	If considered= 5, else= 1
	Data accuracy (A29)	If considered= 5, else= 1
	Data Privacy (A30)	If considered= 5, else= 1
	Data security (A31)	If considered= 5, else= 1
	No of PE (A32)	More than 5 =5, 4–3 =4, 3–2 =3, 2–1 = 2, Not mentioned = 0
	Accuracy (A33)	95–100= 5, 90–94= 4, 89–80= 3, 79–70= 2, > 69= 1, Not mentioned= 0
	Sensitivity (A34)	95–100= 5, 90–94= 4, 89–80= 3, 79–70= 2, > 69= 1, Not mentioned= 0
	Specificity (A35)	95–100= 5, 90–94= 4, 89–80= 3, 79–70= 2, > 69= 1, Not mentioned= 0
	F1 score (A36)	If done= 5, else= 0
	Precision (A37)	If done= 5, else= 0
	ANOVA (A38)	If done= 5, else= 0
	P-value (A39)	> 0.01= 5, not mentioned= 0
	AUC/ AUCROC (A40)	If done= 5, else= 0
	Similarity score (A41)	If done= 5, else= 0
	Recall (A42)	If done= 5, else= 0
	Qualitative bias test (A43)	PROBAST, ROBINS-1, Aquitas etc. if done= 5, if not= 0
	Risk Analysis (44)	If done= 5, else= 0

Data availability Data will be made available by the corresponding author on reasonable request.

Declarations

Conflicts of interests Authors declare that they have no conflicts of interest.

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